

- 5 *Lunularia cruciata* is a primitive plant. Its body consists of a flattened sheet of photosynthetic tissue called a thallus.

Fig. 5.1 shows *L. cruciata* with two different types of reproductive structure, labelled **A** and **B**, on its surface.

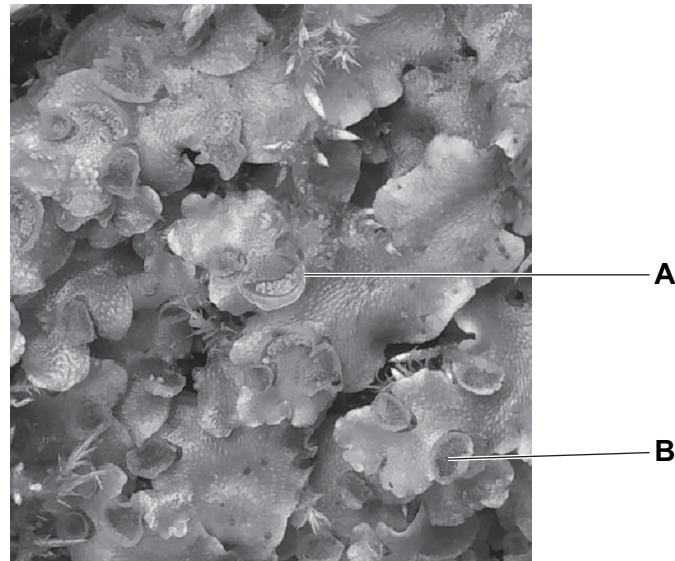


Fig. 5.1

- (a) Structures **A** and **B** and the thallus of *L. cruciata* are haploid.

Explain what is meant by haploid.

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..... [2]

- (b) Structure **A** contains pale discs of tissue, **C**, that can germinate into new *L. cruciata*. The new plants that develop from **C** are genetically identical to the parent plant in Fig. 5.1.

Structure **B** contains male sperm that are chemically attracted to swim to female eggs on a neighbouring parent plant. When the egg and sperm fuse, they form structure **D**. Structure **D** develops to produce spores that grow into new plants that are genetically different from the two parent plants.

Identify which of the structures **A**, **B**, **C** and **D** are:

associated with sexual reproduction

produced by mitosis

the site of meiosis

Each letter may be used once, more than once, or not at all.

[3]

- (c) Fig. 5.2 shows a horse, *Equus caballus*. Horses are diploid animals that reproduce sexually. Male and female horses produce gametes, which fuse to form genetically different offspring.



Fig. 5.2

Explain the need for a reduction division during meiosis in the production of gametes in animals such as horses.

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[Total: 8]